Is the Web App Your AI Built Really Secure?

h5i is a Rust web framework for building apps whose critical rules you can verify, with tools to test the running system.

FRAMEWORK / RUST + AXUM

PROOFS / LEAN 4

TOOLS / DEVELOPMENT INFRASTRUCTURE

Finding Bugs Doesn't Prove Correctness.

Testing and red-teaming can uncover vulnerabilities, but cannot establish that critical properties always hold.

TESTING & RED-TEAMING

Find Bugs ∃

Find vulnerabilities in tested executions; their absence is not a guarantee.

FORMAL VERIFICATION

Prove Properties ∀

Establish stated properties for all modeled inputs, under explicit assumptions.

> Critical rules need stronger evidence: formal verification.

h5i: The Verifiable Web Framework

CORE PRODUCT

Verifiable Web Framework

Build on Axum with composable Rust kernels. Extract application logic to Lean 4 to prove critical rules, with optional GPU acceleration for suitable workloads.

RUST / AXUM | LEAN 4 | OPTIONAL GPU

INTEGRATED TOOLING

Development Infrastructure

A unified CLI for agent-led red-teaming, browser and HTTP automation, security regression tests, and sandboxed workflows.

Works with existing web applications, too.

Write Once. Verify with Lean 4.

Axum and storage stay outside the pure Rust kernel, so application decisions can be extracted and proved. GPU execution is an optional optimization.

Rust / parallel/kernel/src/lib.rs · pure kernel

```rust
pub fn push_rule(event: &[u8], pattern: &[u8],
    display: &[u8], members: u64,
    minimum: u64, maximum: u64, power: u64,
    required: u64, enabled: bool,
    require_display: bool) -> u32 {
    if !enabled { return 0; }
    if power < required { return 0; }
    if members < minimum
        || members > maximum { return 0; }
    if !glob(event, pattern) { return 0; }
    // display-name check omitted
    1
}
```

No I/O and no allocation: Axum and storage adapters stay outside.

Charon
Aeneas

→

Lean 4 / Proofs.lean · verify

```
theorem insufficient_power_denies ...
    (h : power < required) :
    push_rule event pattern display members
      minimum maximum power required true
      requireDisplay = ok 0#u32 := by
  simp [push_rule, h]
```

Rust-CUDA
PTX

→

CUDA / device/src/lib.rs · optional

```rust
#[kernel]
pub unsafe fn push_batch(...) {
    let i = thread::index_1d() as usize;
    ... shared::push_rule(&events[row(i)], ...)
}
```

15.6× 1M push-rule rows vs one CPU thread · RTX 3090

Same kernel source on CPU, in Lean, and on GPU / GPU backend is experimental; Lean does not prove CPU/GPU equivalence

Prove Critical Application Properties

Rust / kernel/src/lib.rs

```rust
fn delete(user: u64, s: &Snapshot,
          id: u64) -> Outcome {
    match find_post(&s.posts, id) {
        None => Err(Error::NotFound),
        Some(p) => {
            if p.author == user
                || is_moderator(&s.moderators, user) {
                Ok((one(Write::DelPost(id)),
                    Reply::Done))
            } else {
                Err(Error::Forbidden)
            }
        }
    }
}
```

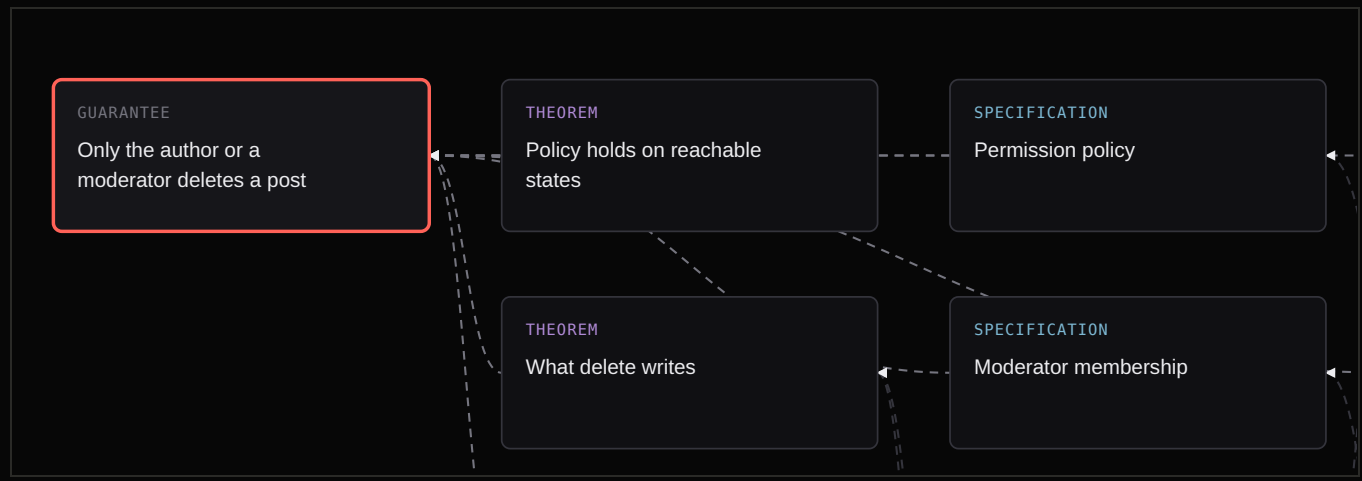
Charon
Aeneas

→

Lean 4 / Theorems.lean · excerpt

```
theorem authorized (a : Principal) (s : Snapshot)
    (c : Command) ws r
    (hinv : Inv (Snapshot.toSt s))
    (h : transition a s c = .ok (.Ok (ws, r))) :
    ∀ w ∈ ws.val, allowed (Snapshot.toSt s) a.user.val w := by -- proof body omitted
```

h5i ui / theorem-spec dependency graph · excerpt



Proofs apply to stated properties under documented assumptions

Development Infrastructure

Red-teaming tests the running system, including HTTP adapters and external services beyond the proof boundary. Untrusted code runs in confined boxes, and findings become CI/CD regression tests.

h5i browser + websec / red-team

```
$ h5i browser snapshot
  @e7 Order 1041
$ h5i websec replay req_3 \
    --set query.id=1042
  → req_4 · HTTP 200
$ h5i websec diff res_3 res_4
  owner: Alice → Bob
```

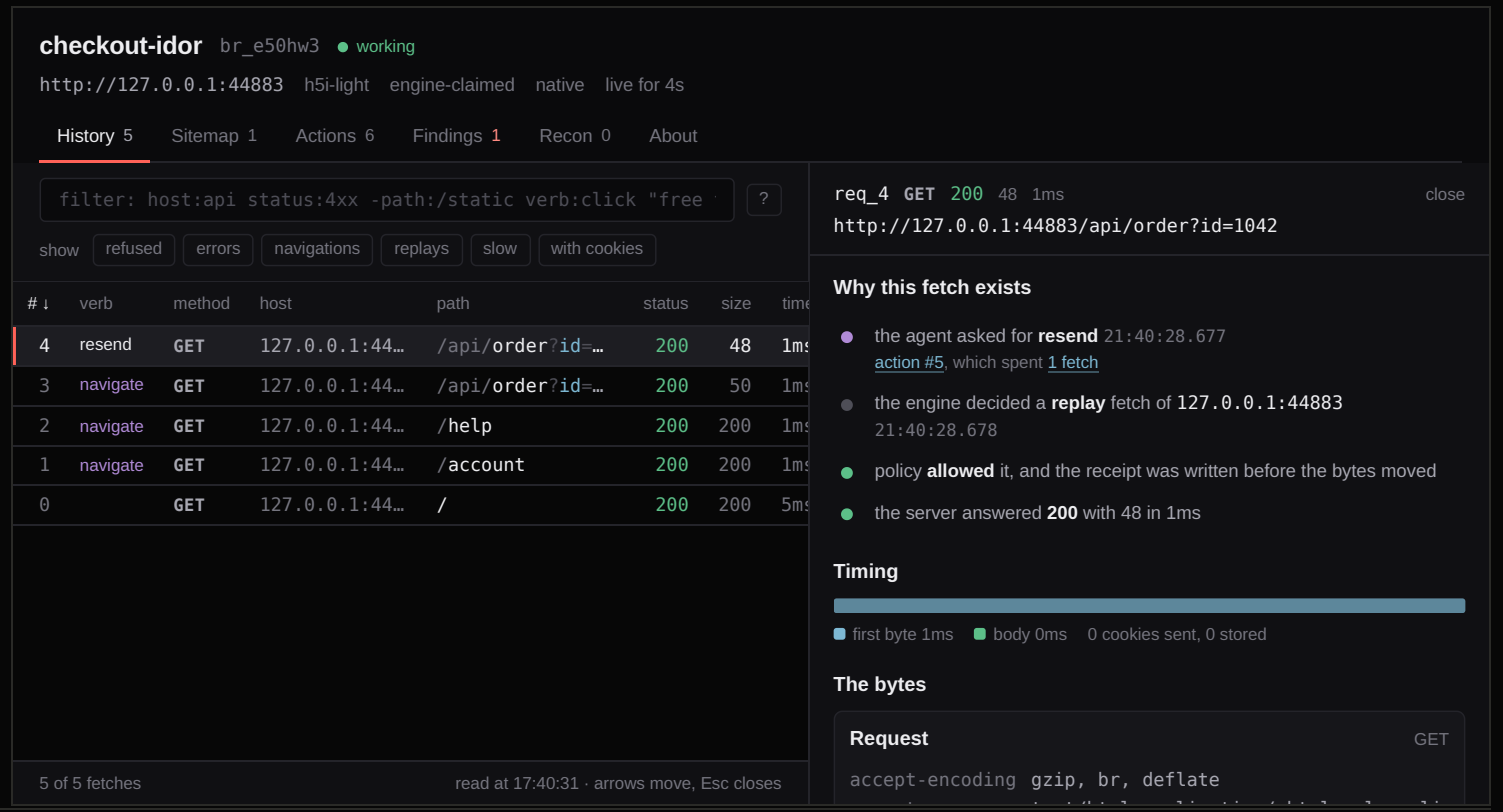
h5i box / sandbox

```
$ h5i box create pr-1234 --pr 1234
$ h5i box run pr-1234 -- npm test
```

h5i test / CI/CD

```
$ h5i test --target $STAGING_URL
  ✔ document-isolation (Pass, 412 ms)
  results: h5i-test-results
```

↔

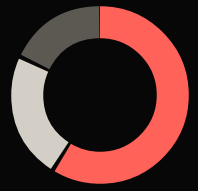


Illustrative shell (browser, websec, box, test) + actual h5i ui / local demo

07 / EARLY TRACTION

Open-source traction

3.5k+



Linux	59%
macOS	23%
Windows	18%

RELEASE DOWNLOADS

880+

GITHUB STARS

Maintainer-reported · downloads November 3, stars October 28, 2026 Repository ↗ Releases ↗

Build and Verify Your Web Apps.

We're looking for developers and teams building security-critical web applications to help shape the next generation of verifiable web development.

Start a conversation ↗

Explore h5i ↗